Rules or Learned Patterns?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Rules or Learned Patterns? before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a **label**led second line to correct your explanation.

Sensors Are Messengers

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

AI or Not?

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Rules or Learned Patterns? trace and record the first step you can explain without help.

Use W2: Complete the supported comparison for Rules and **learned pattern**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Rules or Learned Patterns? explanation against the supplied facts before

> discussing it.

Use the worked case for Rules and **learned pattern**s. Proposed explanation: “A learned rule cannot be written simply”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Fit a new paper classifier

A fictional sorting task has lengths 2, 5, 7, 9 cm with **label**s short, short, long, long. Candidate rules predict long when length is at least 5 or at least 7.

List **prediction**s and errors for both, choose one, and predict a new length of 6 before any **label** is revealed. State one limit of your conclusion.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Rules or Learned Patterns? target to a trace and a justified conclusion.

Sorter A follows 'red left, blue right'. Sorter B was trained using **label**led red and blue cards. Both send a purple card left. Does their shared **output** show that they used the same **process**? What **evidence** is missing?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Rules or Learned Patterns? case and show the deciding evidence.

Sorter A follows a written red-card-left rule. Sorter B is fitted using **label**led leaves.

How were their decision **process**es obtained? Where should A send blue cards?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

System A uses the **written rule** green means go. System B learns fruit **label**s from example photographs. Who supplies A's rule? What changes when B is trained on different examples?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 4 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Rules or Learned Patterns? adds to Helpful Classroom Sorter and keep the evidence.

Draw a fixed-rule version and a learned-pattern version of the Classroom Sorter; state what **evidence** would justify the learned version.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Sorter A follows 'red left, blue right'. Sorter B was trained using **label**led red and blue cards. Both send a purple card left. Does their shared **output** show that they used the same **process**? What **evidence** is missing?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Execute before fitting

Execute before fitting. Apply the supplied Rules or Learned Patterns? facts and keep your reasoning.

Use wide when width is at least 5 cm. Predict the **label** for widths 4, 5 and 6. Then state what changes if the rule uses greater than 5 instead.

My reasoning and evidence

W2 Complete the supported comparison

> Complete the supported comparison. Apply the supplied Rules or Learned Patterns? facts and keep your reasoning.

Widths 1, 3, 5, 7 have **label**s narrow, narrow, wide, wide. For candidates 3 and 5, list all **prediction**s using wide when width is at least the candidate. Count disagreements and select the lower-error candidate.

> **My reasoning and evidence**

W3 Make a tie visible

> Make a tie visible. Apply the supplied Rules or Learned Patterns? facts and keep your reasoning.

For the worked widths 2, 4, 6, 8 with **label**s narrow, narrow, wide, wide, compare **threshold**s 5 and 6. Give a width where they predict differently. Explain why the four **fitting** examples cannot decide that case.

> **My reasoning and evidence**

WORKBOOK · 15 OF 21

W4 Fit a new paper classifier

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Fit a new paper classifier. Apply the supplied Rules or Learned Patterns? facts and keep your reasoning.

A fictional sorting task has lengths 2, 5, 7, 9 cm with **label**s short, short, long, long. Candidate rules predict long when length is at least 5 or at least 7.

List **prediction**s and errors for both, choose one, and predict a new length of 6 before any **label** is revealed. State one limit of your conclusion.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Keep the check honest

> Keep the check honest. Apply the supplied Rules or Learned Patterns? facts and keep your reasoning.

A learner sees that their final-check card was predicted incorrectly, changes the **threshold** to match it and reports the same card as an unseen test. Explain the problem and propose a valid next check.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Rules or Learned Patterns?

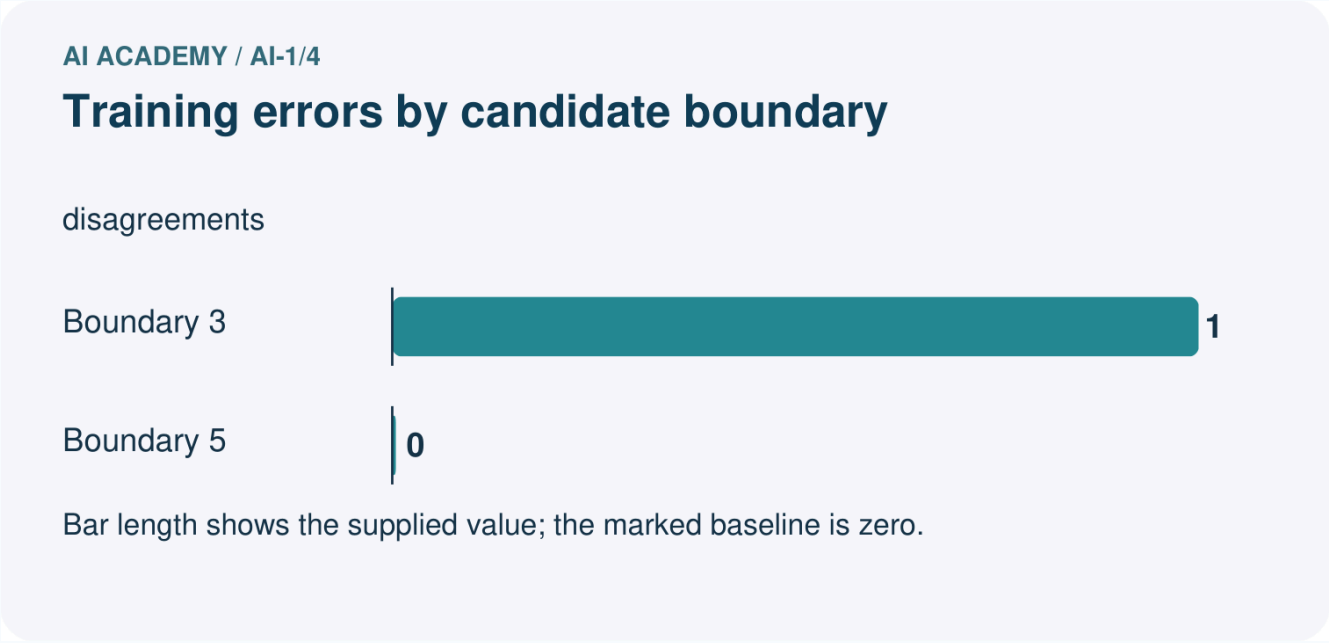
> A fictional paper sorter learns a size boundary for toy pebbles. Explain the labelled step

> before checking the changed condition.

A fictional paper sorter learns a size boundary for toy pebbles.

Training diameters are 1,3,5,7 cm with **label**s small,small,large,large.

Compare large-at-least-3 with large-at-least-5; choose fewer errors.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/4

Rules or Learned Patterns?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training examples

Diameters 1,3,5,7 have reference labels S,S,L,L.



2. Candidate 3

Predictions S,L,L,L: one disagreement at 3.



3. Candidate 5

Predictions S,S,L,L: zero disagreements.



4. New pebble

Selected boundary 5 predicts small for a new diameter 4.

CHANGE ONE THING

Give a new pebble diameter 5 instead of 4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Show which stage used **label**s and which stage used only the new measurement.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Document whether the sorter's rules were teacher-written or chosen from examples, retaining candidate errors.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Document whether the sorter's rules were teacher-written or chosen from examples, retaining candidate errors.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper sorter learns a size boundary for toy pebbles.

Practice 1: explain the deciding evidence

Which **label**led step matches this detail? Selected boundary 5 predicts small for a new diameter 4.

1. **Training** examples

2. Candidate 3

3. Candidate 5

4. New pebble

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: New pebble Selected boundary 5 predicts small for a new diameter 4. The learning procedure compares candidate rules against supplied examples and selects boundary 5.

Applying that boundary to diameter 4 is a later **prediction** step. An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give a new pebble diameter 5 instead of 4. Which conclusion is supported by the supplied facts?

1. The selected rule predicts large.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The selected rule predicts large. At least includes equality, so 5 meets the boundary.

Repair a missing building block

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

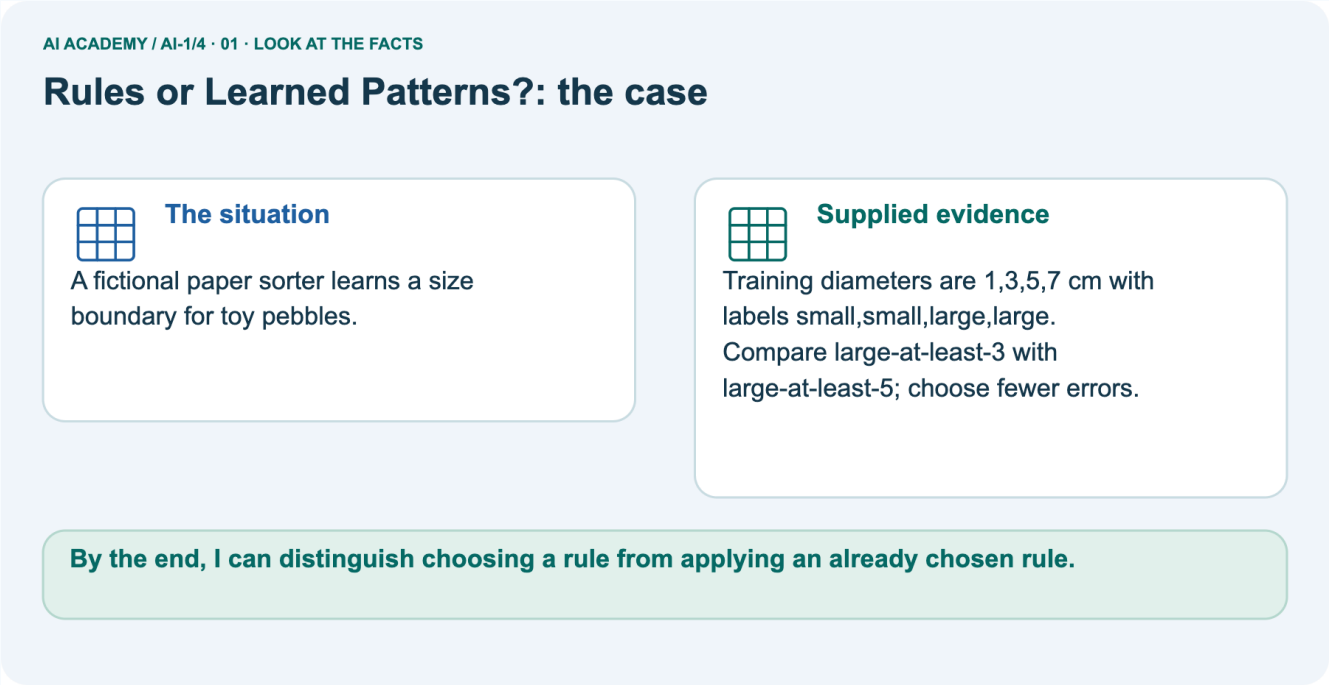
> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Rules or Learned Patterns? picture from its labels, then record a relevant limit.

01. Rules or Learned Patterns?: the case



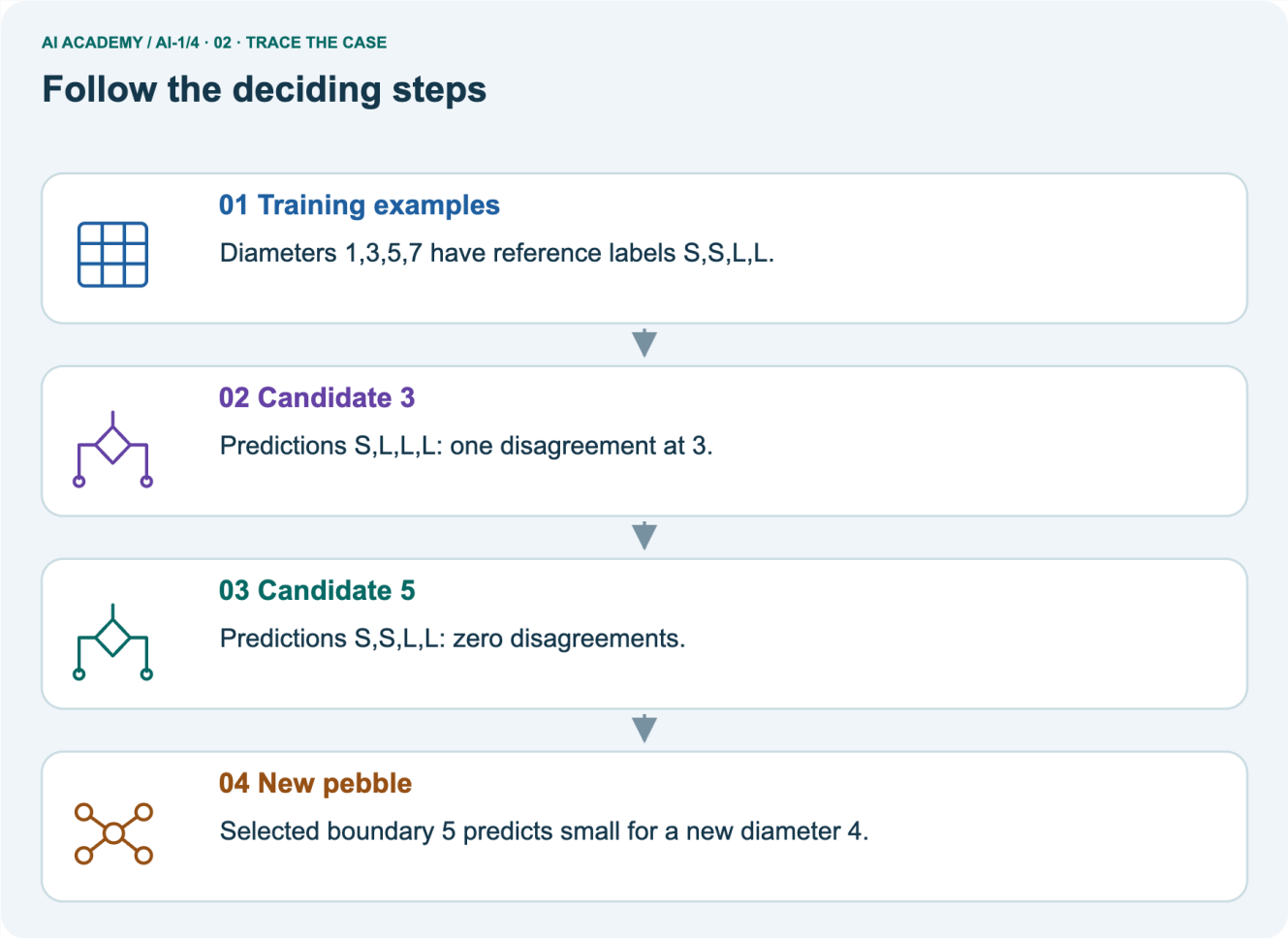
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper sorter learns a size boundary for toy pebbles. Supplied facts: **Training** diameters are 1,3,5,7 cm with **label**s small,small,large,large. Compare large-at-least-3 with large-at-least-5; choose fewer errors.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



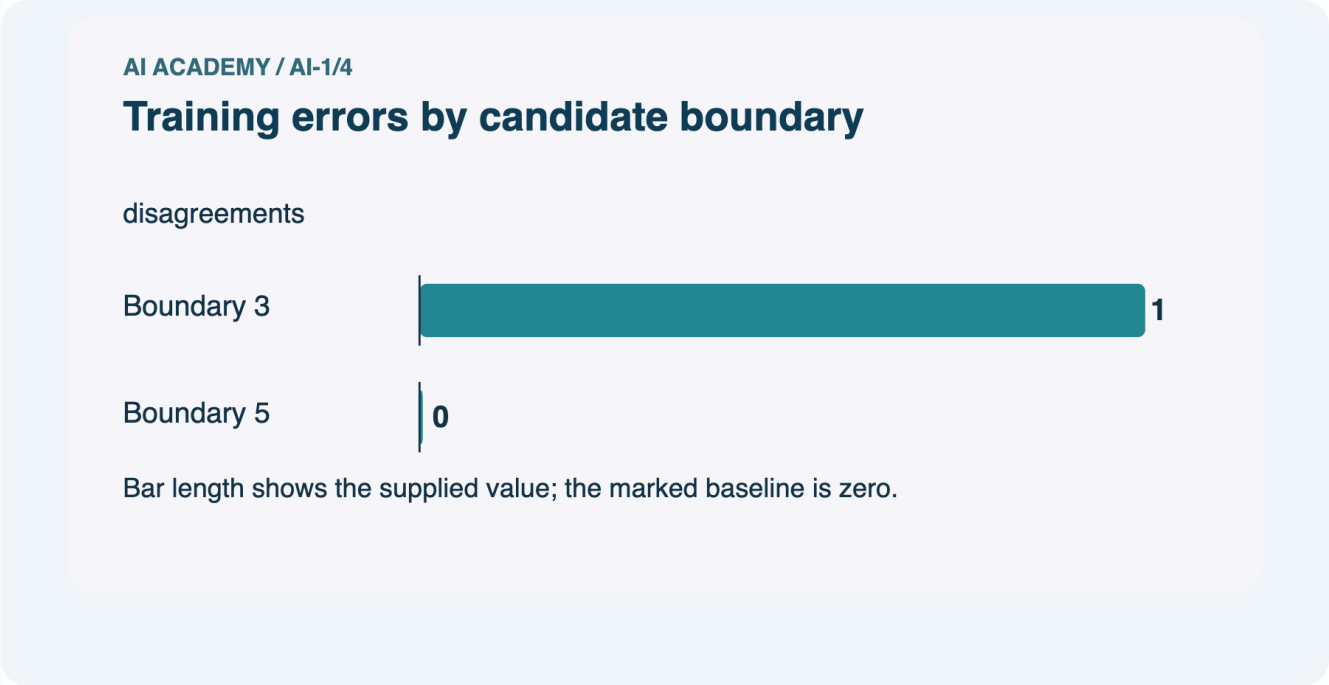
Follow the labelled picture with the matching explanation. Open the vector diagram

Training examples: Diameters 1,3,5,7 have reference **label**s S,S,L,L. → Candidate 3: **Prediction**s S,L,L,L: one disagreement at 3. → Candidate 5: **Prediction**s S,S,L,L: zero disagreements. → New pebble: Selected boundary 5 predicts small for a new diameter 4.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Training errors by candidate boundary



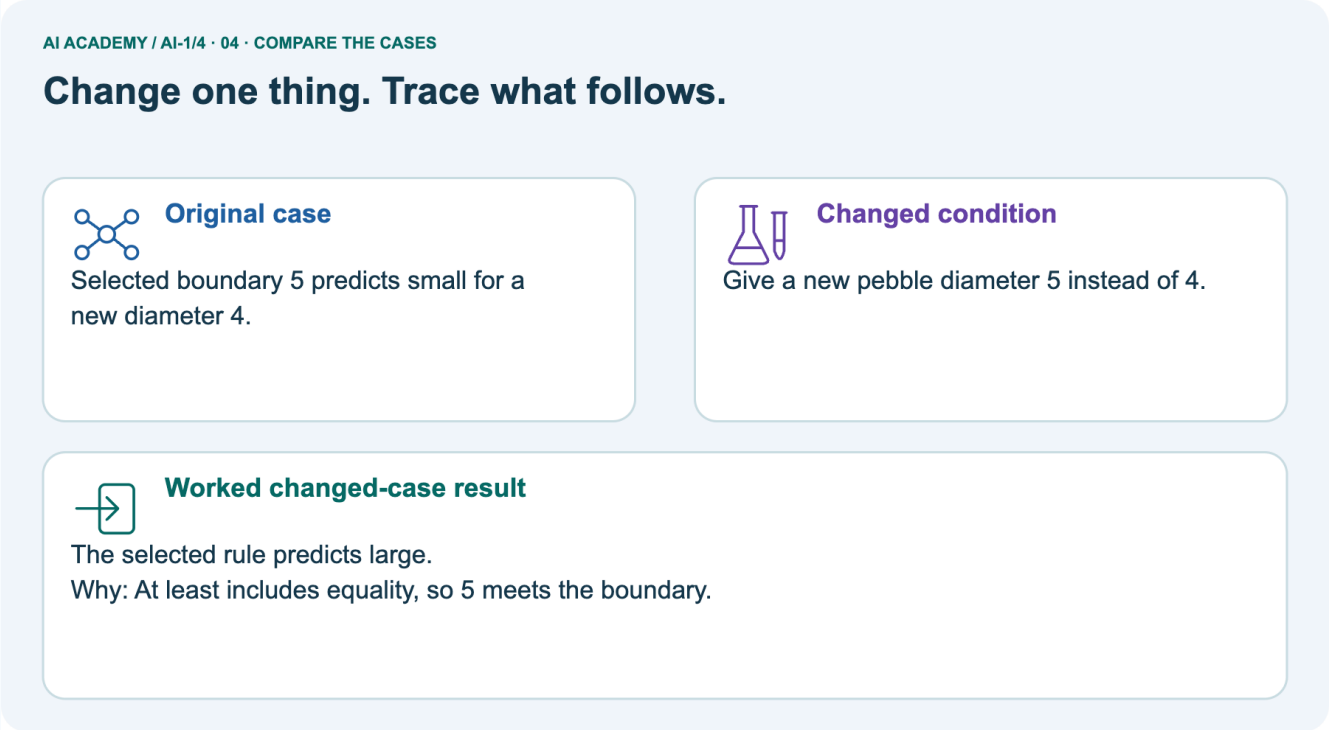
Follow the labelled picture with the matching explanation. Open the vector diagram

Training errors by candidate boundary. Units: disagreements. Boundary 3: 1. Boundary 5: 0.

Read the **label**s and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

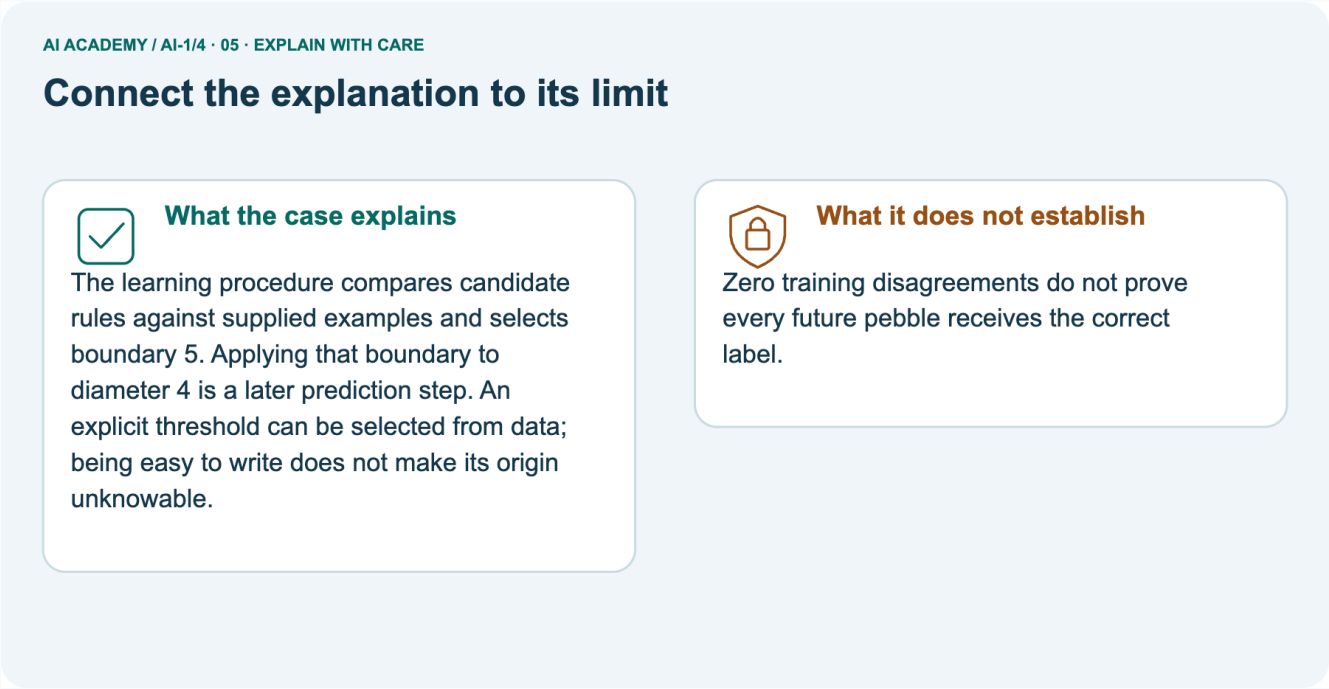
Original case: Selected boundary 5 predicts small for a new diameter 4. Changed condition: Give a new pebble diameter 5 instead of 4. Expected result: The selected rule predicts large.

Reason: At least includes equality, so 5 meets the boundary.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

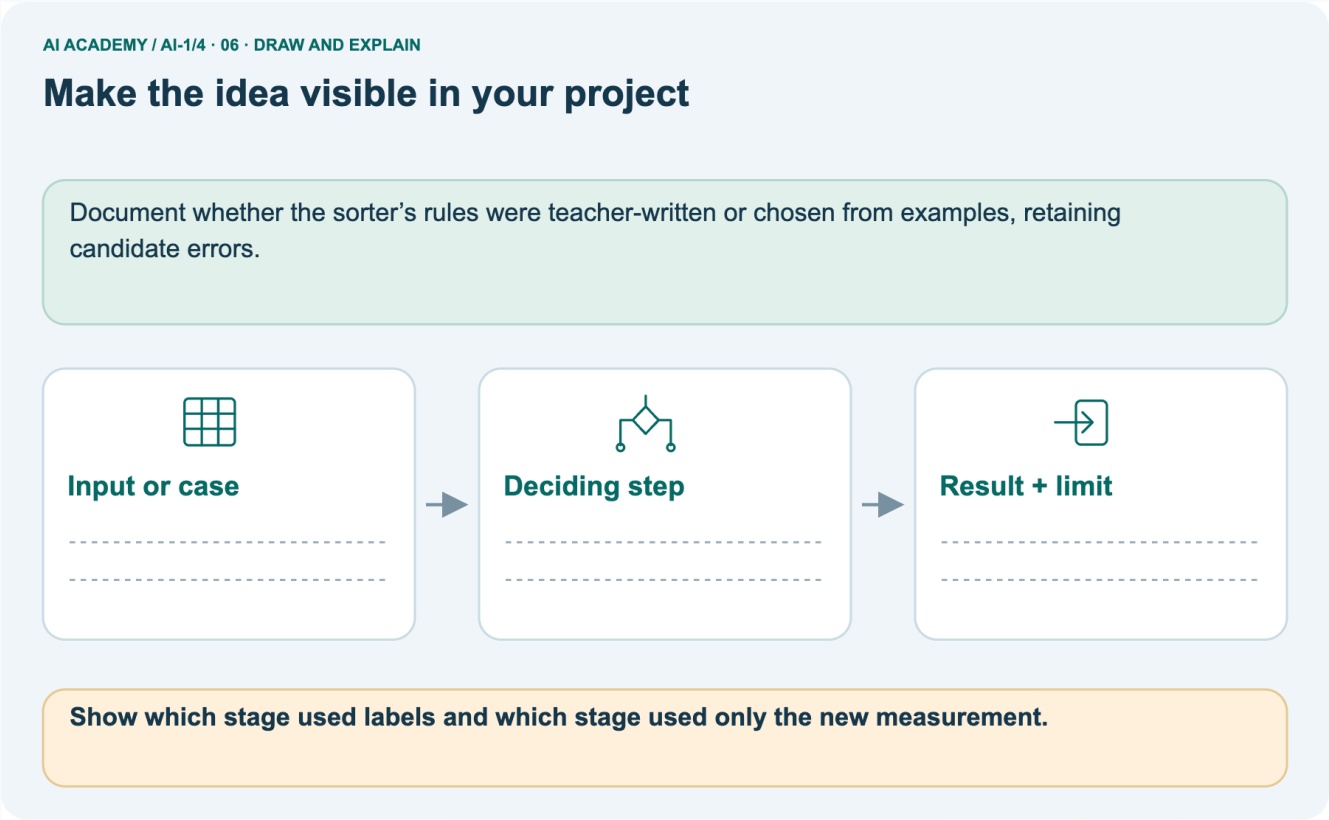
The learning procedure compares candidate rules against supplied examples and selects boundary 5. Applying that boundary to diameter 4 is a later **prediction** step.

An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable. Limit: Zero **training** disagreements do not prove every future pebble receives the correct **label**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document whether the sorter's rules were teacher-written or chosen from examples, retaining candidate errors. Draw your **input**, deciding step, result and limit.

Supported question: Show which stage used **label**s and which stage used only the new measurement.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.